Machine Learning Regression on Synthetic Gravitational-Wave Inspirals

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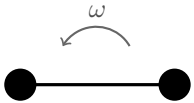
Motivation

- Gravitational-wave (GW) astronomy relies on recovering source parameters from noisy detector data
- Classical parameter estimation (e.g. LIGO's matched filtering + Bayesian inference) is **computationally expensive**
- **Goal:** Build an ML surrogate model that maps GW inspiral features to source parameters

Quadrupole Approximation

- A **quadrupole moment** describes the asymmetric shape of a mass distribution
- This changing configuration **radiates gravitational waves** (Creighton 2026)



Rotating dumbbell

Analogy: the shape of a dumbbell changes as it rotates or evolves

Key Assumptions

- Weak-field regime
- Non-relativistic velocities ($v \ll c$)
- System size $\ll$ gravitational wavelength

Physical Parameters

System Parameters

- Masses: $m_1, m_2 \in [1.2, 50]\, M_\odot$
- Initial Separation: $r_0 \in [5 \times 10^6, 10^8]$ km
- Distance to observer: $d \in [10, 10^5]$ kpc

Physical Constants

- $G = 6.674 \times 10^{-11}\ \mathrm{m}^3\ \mathrm{kg}^{-1}\ \mathrm{s}^{-2}$
- $c = 3 \times 10^8$ m/s
- $M_\odot = 1.989 \times 10^{30}$ kg

- Circular orbits assumed ($e \approx 0$), consistent with compact binaries (Peters 1964)
- **1,000 synthetic samples** generated in total

Orbital Decay & Frequency

Orbital Decay (Peters 1964)

$$\frac{dr}{dt} = -\frac{64}{5}\frac{G^3 m_1 m_2 (m_1 + m_2)}{c^5 r^3}$$

- Solved numerically via `scipy` RK45
- Tracks how separation shrinks with time

Orbital Frequency (derived in the report)

$$f_{\text{orb}} = \frac{1}{2\pi}\sqrt{\frac{G(m_1 + m_2)}{r^3}}$$

- Derived from $F_g = F_{\text{cen}}$
- Increases as separation decreases ("chirp")

Strain Amplitude & Target Labels

GW Strain (Creighton 2026)

$$h(t) = \frac{4(G\mathcal{M})^{5/3}\,\omega_r^{2/3}}{c^4\,d}$$

- $\mathcal{M}$: chirp mass
- $\omega_r = 2\pi f_{\text{orb}}$: orbital angular frequency
- d: distance to observer

Target Labels

Chirp mass:

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$$

Merger time:

$$t_m = \frac{5}{256}\frac{c^5 r_0^4}{G^3 m_1 m_2 (m_1 + m_2)}$$

Dataset & Feature Engineering

20 Input Features

- $h_{\text{noisy}}(t_1), \ldots, h_{\text{noisy}}(t_{17})$ (noisy strain time-series)
- d: observer distance
- r_0: initial separation
- $\mu = \frac{m_1 m_2}{m_1 + m_2}$: reduced mass

Preprocessing

- No missing values (synthetic data)
- **Log-transform** applied — large SI-unit magnitudes
- Values converted back to linear space for MAE
- **75/25 train-test split** with fixed random seed

Why Random Forest?

- **Ensemble method:** combines many decision trees for better generalization (Kavlakoglu 2025)
- **Non-linearity:** strain $\rightarrow$ chirp mass / merger time mapping is highly non-linear
- **Scale-invariance:** features span vastly different physical scales and RF handles this naturally
- **Overfitting mitigation:** ensemble averaging reduces variance with only 1,000 samples
- Implementation: `scikit-learn` `RandomForestRegressor`

Experiment 1 — Full Feature Set

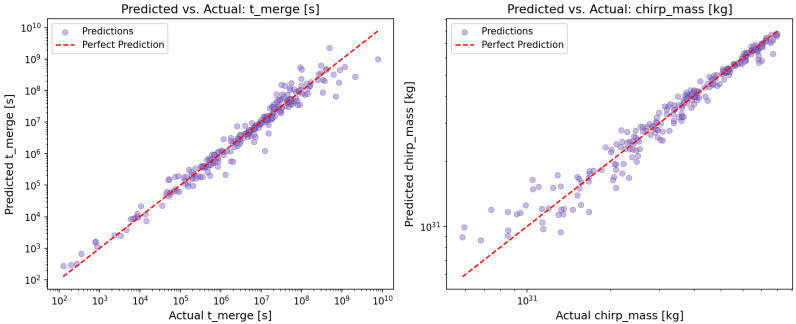


Figure 1: Predicted vs. Actual — full feature set

- Decent training fit but poor generalization
- Model **memories** r_0 **values**; struggles at extremes
- Large MAE reflects SI-unit scale

	Train	**Test**
MAE	4.1×10^{29}	1.1×10^{30}
R^2	0.87	0.59

Feature Importance — Full Feature Set

- r_0 **dominates** ($\sim$75%) for merger time prediction and μ **is second** ($\sim$25%) for chirp mass prediction
- Both appear directly in the *analytical formulae*:
  - $t_m \propto r_0^4$
  - $\mathcal{M}$ depends on μ
- Strain features contribute **effectively nothing**
- Model learned **algebra, not physics**

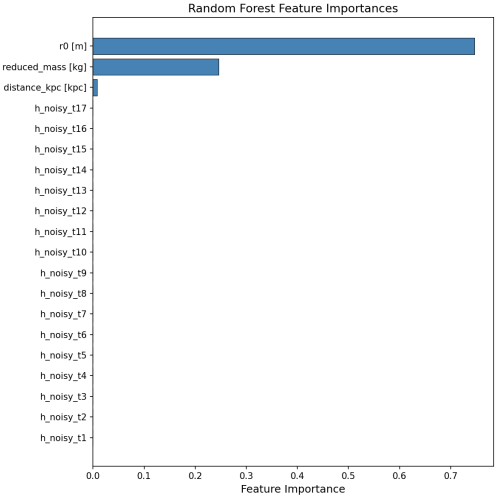


Figure 3b: Feature importances — full feature set

Experiment 2 — Strain-Only Features

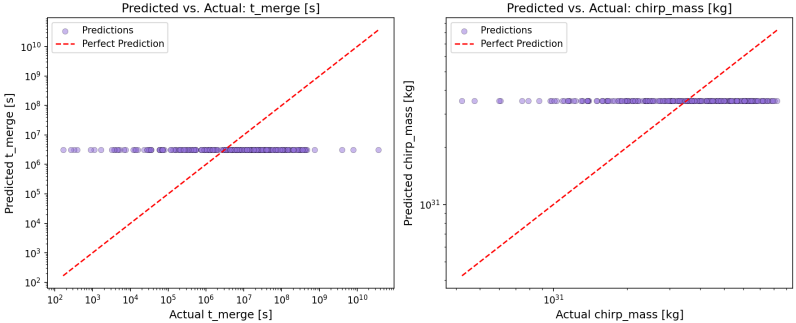


Figure 2: Predicted vs. Actual — strain-only features

- Removed r_0, d, μ to test whether strain alone contains learnable signal
- Increased strain samples from 17 to **20 time-stamps**
- Predictions collapse to a *flat horizontal line* indicating the dataset mean
- Classic signature of a model that **found no learnable signal**

Feature Importance — Strain-Only

- All feature importances are effectively **zero**
- Apparent negative values are a *plotting artifact* of axis scaling
- Model cannot extract any meaningful signal from 20 noisy strain snapshots
- Confirms: raw strain time-series is **insufficient** for this regressor

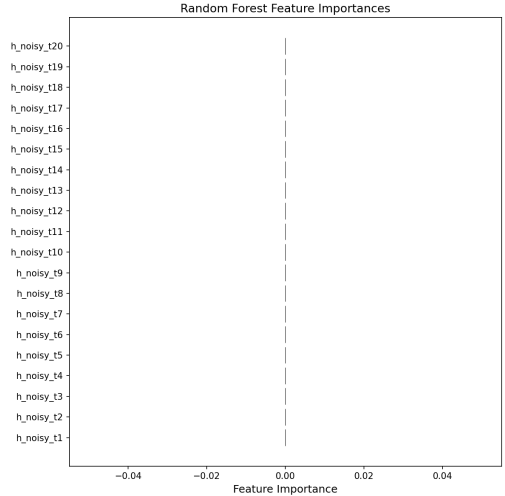


Figure 3a: Feature importances — strain-only

Conclusion

Key Findings

- A Random Forest with **full features** ($R^2 = 0.87$ train) was exploiting algebraic shortcuts, not learning from strain
- Restricting to **strain only** caused complete model failure ($R^2 \approx 0$)
- Real GW parameter estimation uses **matched filtering** + Bayesian inference (LALInference (Veitch et al. 2015), Bilby (Ashton et al. 2019)) — far beyond a simple regressor

Lessons Learned

- Including analytically-derived features *leaks* the answer to the model
- 17–20 noisy time samples are **too sparse** to capture waveform morphology
- Feature importance is a vital diagnostic tool

Different Approach

- Generate a **denser strain time-series** (hundreds of time steps spanning the full inspiral)
- **Normalise strain by distance** to remove d as an implicit shortcut
- Explore **other model architectures** that can extract waveform morphology
- Compare against LIGO's **matched filtering** pipeline as a baseline
- Use **Bayesian inference** frameworks (Bilby) to produce posterior distributions over parameters

References I

- Ashton, G. et al. (2019). "Bilby: A User-Friendly Bayesian Inference Library for Gravitational-Wave Astronomy". In: *The Astrophysical Journal Supplement Series* 241, p. 27.
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